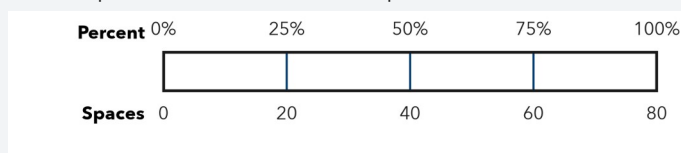


LESSON 4-3 • SESSION 1 • EXPLAIN YOUR THINKING**Estimate with Benchmarks**

Study the model. Then solve your own problem the same way. Use only the help you need.

WORDS estimate (*estimar*) • benchmark percent (*porcentaje de referencia*) • the whole (*el entero*)**1 • SEE A MODEL***read it with a partner***SEE A MODEL** *a problem like yours, solved*

A parking lot has 80 spaces. 61 spaces are full. About what percent of the lot is full?

**HOW I SOLVED IT**

- 1 Name the whole.** 80 spaces = 100%.
- 2 Find benchmarks.** 25% = 20, 50% = 40, 75% = 60.
- 3 Compare.** 61 is very close to 60, so the lot is about **75%** full.

I SAY

"The whole is **80**. The part is close to **60**, which is **75%** of the whole."

I WRITE

I know 61 out of 80 is about **75%** because **75%** of **80** is **60**, and 61 is close to 60.

2 • DRAW IT*your turn: label the whole, the part and the percent*

A water tank holds 120 gallons. It has 82 gallons in it. About what percent of the tank is full? Use the bar to find benchmark amounts.

**TRY THESE STEPS** *fill in the blanks as you work*

- 1** Name the whole: _____ gallons = 100%.
- 2** Fill in the benchmarks: 25% = _____, 50% = _____, 75% = _____.
- 3** Which benchmark is 82 closest to?

3 • SAY IT*tell a partner*

Partner A: "The whole is _____. The part is close to _____, which is _____% of the whole." Partner B: "How do you know it is closer to _____% than to _____%?"

4 • WRITE IT*pick a starter • name the whole • use units***WORD BANK** about • benchmark • close to • between • the whole • estimate

Because I know _____ out of _____ is about _____% because _____% of _____ is _____.

But Some students think 82 out of 120 is 82%, but the whole is _____, so _____.

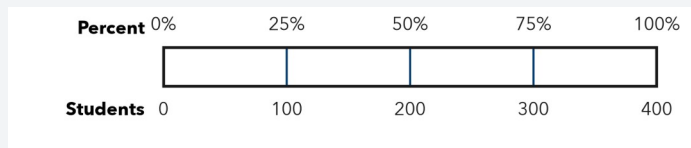
So _____ is between _____ and _____, so the percent is between _____% and _____%.

LESSON 4-3 • SESSION 2 • EXPLAIN YOUR THINKING**Estimate a Percent of a Number**

Study the model. Then solve your own problem the same way. Use only the help you need.

WORDS *estimate (estimación)* · **compatible (friendly) numbers** (*números compatibles*) · **ten percent** (*diez por ciento*)**1 • SEE A MODEL***read it with a partner***SEE A MODEL** *a problem like yours, solved*

A school has 396 students. About 24% walk to school. Estimate the number of students who walk, two ways.

**HOW I SOLVED IT**

- 1 Way 1.** $396 \approx 400$ and $24\% \approx 25\%$. 25% is $\frac{1}{4}$, so $400 \div 4 =$ **about 100**.
- 2 Way 2.** $396 \approx 400$ and $24\% \approx 20\%$. 10% of $400 = 40$, so $20\% =$ **about 80**.
- 3 Compare.** 24% is much closer to 25% than to 20%, so **100 is the better estimate**.

I SAY

"I used **25%** of **400** because **400 is easy to divide by 4**." "I used **20%** of **400**. Our estimates are different because **we rounded the percent differently**."

I WRITE

I rounded **396** to **400** because **400** is easy to divide by **4**. 24% is close to 25%, so **about 100 students walk**.

2 • DRAW IT*your turn: label the whole, the part and the percent*

A theater sold 588 tickets. About 26% were for kids. Estimate the number of kids' tickets two ways, using different compatible (friendly) numbers.

**TRY THESE STEPS** *fill in the blanks as you work*

- 1** Way 1: round 588 to _____ and 26% to _____%.
- 2** Way 2: round 588 to a different friendly number: _____.
- 3** Which estimate is closer? Why?

3 • SAY IT*tell a partner*

Partner A: "I used _____% of _____ because _____." Partner B: "I used _____% of _____." Our estimates are different because _____."

4 • WRITE IT*pick a starter · name the whole · use units***WORD BANK** round · compatible (friendly) number · about · estimate · closer · divide by 4**Because** I rounded _____ to _____ because _____ is easy to divide by _____.**But** Some students think only one estimate is right, but _____ and _____ are both close because _____.**So** I rounded _____ up, so my estimate is a little _____.